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Shipboard paleomagnetic age estimates for an acoustic basement emplacement in Marion Plateau, off northeast Australia

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Ocean Drilling Program Leg 194 was carried out in the Marion Plateau, located to the northeast of Australia. Drilling penetrated few metres of altered olivine basalt flows, underlying the sediments. Shipboard paleomagnetic investigations on these basalts resulted in well-defined paleomagnetic directions. The directions from the archive-half long-core measurements at hole 1193C between 540.97 and 543.97 m below seafloor, together with the archive-half long-core and discrete measurements from the working-half of hole 1198B between 513.85 and 517.25 m below seafloor, yielded an overall mean inclination value of 52.7° ($\alpha_{95} = 4.0°$, $K = 537$, $n = 129$, $N = 4$). These give the paleolatitude of the Marion Plateau to be 33.3°S with upper and lower 95% confidence limit 37.3°S and 29.6°S, respectively. Comparison of the corresponding paleo-colatitude result with the Australian apparent polar wander path indicates a possible emplacement time for the basalt of either 130–110 Ma or 190–165 Ma. The latter result is in better agreement with the available $^{40}Ar/^{39}Ar$ age of 162 $\pm$ 0.9 Ma reported for the basalt.

KEY WORDS: acoustic basement, Marion Plateau, ODP Leg 194, paleomagnetism.

INTRODUCTION

The Marion Plateau is located south of the Queensland Plateau in the Great Barrier Reef Marine Park of Australia, between latitudes of 19° and 22°S and longitudes 151° and 154°E. It is believed to represent the offshore extension of the Queensland Plateau to the south of the Townsville Trough (Figure 1), a present-day bathymetric feature separating the two plateaus.

Previous research in the eastern Coral Sea basin had identified two distinct tectonic episodes. A Late Jurassic–Early Cretaceous extensional event, which formed the Queensland and Townsville Troughs, and the Late Cretaceous rifting, which created numerous continental fragments now capped by carbonate platforms such as the Marion and Queensland Plateaus (Struckmeyer & Symonds 1997).

Prior to the Ocean Drilling Program (ODP) Leg 194 drilling, the timing of basin formation and the age of the sedimentary succession of the Townsville Basin were interpreted within a regional tectonostratigraphic context, as there were no direct controls on its stratigraphy. As a consequence, it was assumed that the basement of the Marion Plateau would be similar to that of the Queensland Plateau, which was cored during ODP Leg 133 (Sites 824 and 825: Figure 1) and determined to be fine-grained dark-grey, poorly foliated well-lithified quartz–feldspar–mafic metasediment or metavolcanic rocks. The seismic investigation of the area clearly identified the boundary of the Marion Plateau and the position of the strong seismic reflector coincides with the interface between the unconsolidated siliciclastic sediment and the basal consolidated bedrock. This bedrock, known to be a strong seismic reflector, is termed the acoustic basement and was penetrated at four sites during Leg 194. The acoustic basement rocks collected during this leg are altered volcanic flows and volcaniclastics, which differ greatly from the metasedimentary rocks found on the Queensland Plateau.

During Leg 194 drilling, holes intersected the Holocene to Late Miocene sedimentary rocks of the Marion Plateau and also penetrated into the acoustic basement at four holes (1193C, 1194B, 1197B, 1198B) at different sites. The recovered acoustic basement was found to be unmetamorphosed altered olivine basalt and volcaniclastic breccia, thus making it different from that of the basement rocks at the Queensland Plateau.

As part of a multidisciplinary development of a chronostratigraphic framework routine paleomagnetic measurements were performed on archive-half long-core samples. It was also of considerable interest to have an age estimate for the basalt basement while Leg 194 was still under way in an effort to constrain the time of basin formation in the Marion Plateau. To this effect, we

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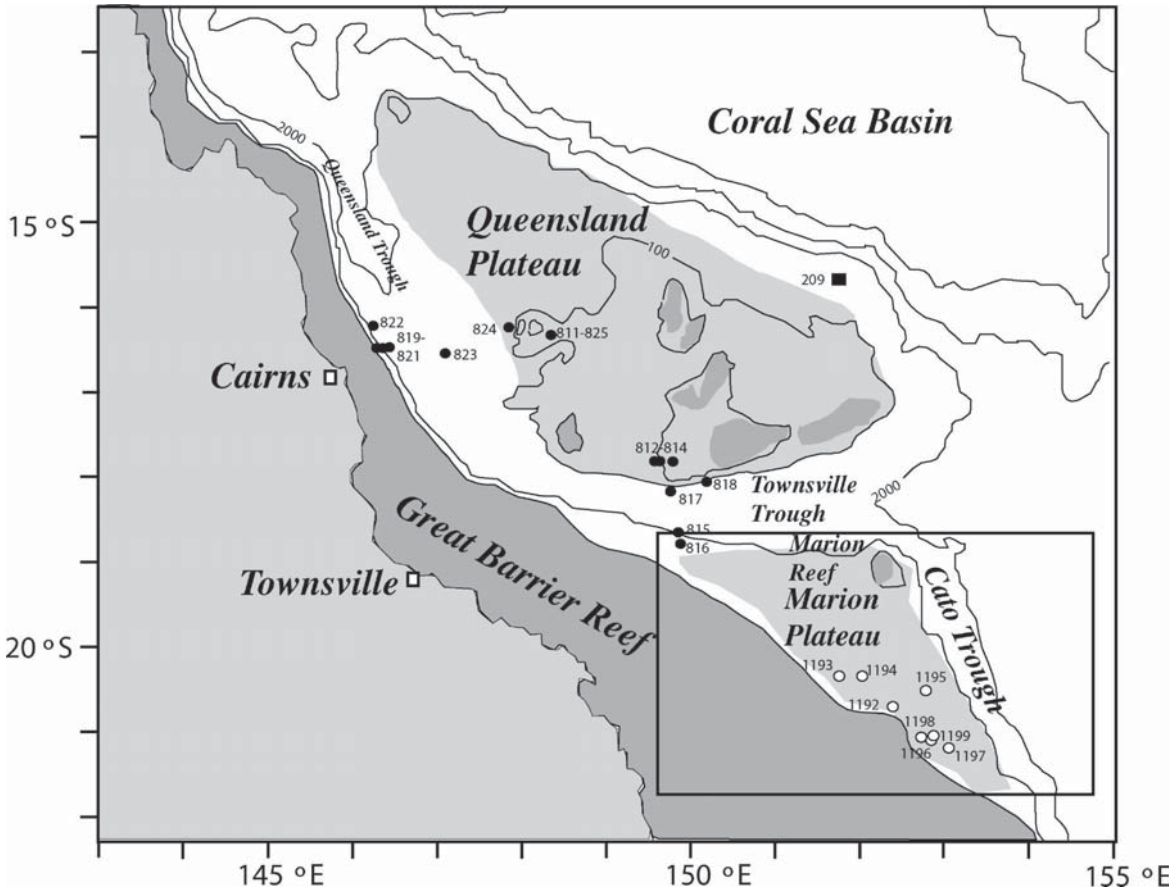


Figure 1 (a) Location map of the Marion Plateau and Queensland Plateau, offshore northeastern Australia: ODP Leg 194 sites are shown in the rectangle.

also made discrete sample measurements from the working-half of the long cores in order to compare results. The recovered core fragments were neither oriented with respect to magnetic north, nor with each other. Consequently, only magnetic inclination data were used. Fortunately, the recovered basalt had a magnetisation signal strong enough to give a meaningful result on board the ship. After the routine paleomagnetic measurements and data processing, the inclinations of primary and secondary magnetisations carried by the magnetic minerals in the units were used to calculate paleolatitudes. Finally, by plotting and fitting these results to the Australian apparent polar wander path, approximate ages could be determined. While this is very much a preliminary age determination, given the small amount of material recovered, the age of emplacement determined onboard the ship was later confirmed by radiometric age determination carried out subsequently.

PALEOMAGNETISM

Natural remanent magnetisation (NRM)

The acoustic basement, represented by olivine basalt, was encountered in holes 1193C at 539 mbsf (metres below sea-level), 1194B at 420 mbsf, and 1198B at 500 mbsf, and a volcaniclastic breccia was recovered from hole 1197B at 656 mbsf. The sediment to volcanics transition was marked by abrupt increase in the NRM intensity from 10^{-5} to 10^{-1} and 10^{-2} A/m, clearly indicating the position of the relatively strongly magnetised basalt. However, comparison of this NRM intensity value with the global average intensity value by Prévot & Grommé (1975), for subaerial (3.1–4.1 A/m) and submarine basalt (1.7–7.4 A/m) shows that the intensities obtained here are about an order of magnitude less, which could have resulted from alteration or low content of iron oxides.

Continuous long-core measurements were carried out routinely at all sites where the basalts were recovered. Directional checks were also performed on discrete samples collected from two representative holes where rocks were not fractured and collection of discrete samples was possible. The archive-half long-core samples were subjected to stepwise alternating field (AF) demagnetisation, mostly up to 30mT: this is the step at which most of the magnetic overprints are believed to disappear without completely demagnetising the archive-half, in keeping with ODP policy. The results for each hole are given below.

Paleomagnetic directions

HOLE 1193C

In hole 1193C, 4.5 m-thick basalt was recovered at 539–544 mbsf and its magnetisation was then measured as a continuous long-core sample. The results of this measurement, the inclination and magnetisation intensity at NRM and after 30mT demagnetisation, are shown in Figure 2. A decrease of one- to two-orders of magnitude of the NRM intensity are observed in the depth intervals 538.95–539.15, 540.00–540.10 and 540.56–540.91 mbsf (shaded areas in Figure 2a) with the demagnetised intensities approaching zero. The first intensity low in the uppermost part of the section (i.e. 538.95–539.15 mbsf) is related to the sediment portion of the core. On the other hand, the two lower intensity lows (i.e. at 540.00–540.10 and 540.56–540.91 mbsf) correspond to anomalous inclination values, suggesting possible excursions or short-term geomagnetic events (Figure 2a). The relative NRM intensity values above and below these two anomalous intervals are stronger and stable. These two stable relative NRM intensity values separated by the two abnormal intervals can be identified in Figure 2a: the first between 539.15 and 539.95 mbsf (average NRM intensity = 0.62 A/m, $\sigma = 0.34$, $N = 17$) and the second between 540.96 and 543.97 mbsf (average NRM intensity = 0.34 A/m, $\sigma = 0.14$, $N = 58$).

The inclinations in the stable intervals before demagnetisation for the top and bottom intervals are statistically similar [i.e. the mean for the top stable interval would be –39.6° ($\alpha_{95} = 5.8°$, $N = 16$, $K = 40.8$, excluding one anomalous point) and –43.8° ($\alpha_{95} = 1.7°$, $N = 58$, $K = 126.2$) for the bottom interval]. Because these inclinations are statistically indistinguishable averaging the two, gives a value of –42.9° ($N = 74$, $K = 85.8$). However, the average inclination after 30 mT demagnetisation, for the upper and lower stable intervals with mean inclinations values of –41.7° ($\alpha_{95} = 6.1°$, $N = 16$, $K = 38.1$) and –57.5° ($\alpha_{95} = 1.6°$, $N = 58$, $K = 143.9$), respectively, are distinctly different (Figure 2a).

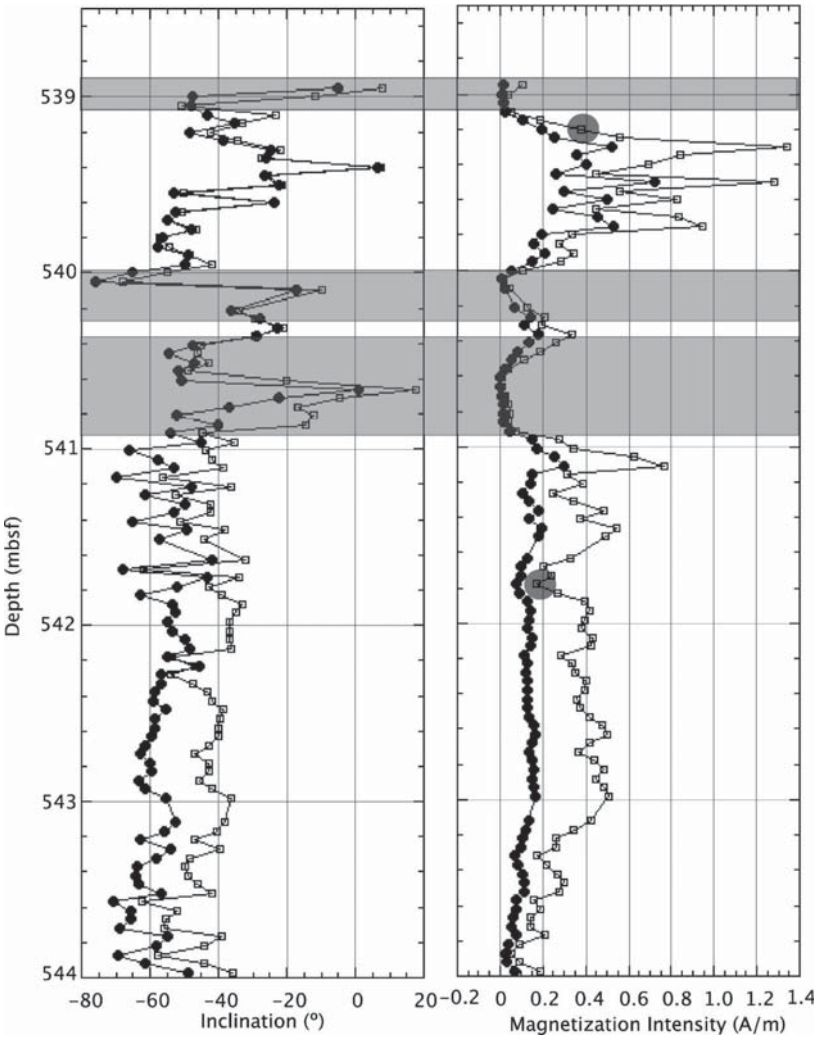


Figure 2 (a) Archive-half measurements on acoustic basement in hole 1193C showing inclination and magnetisation intensity values at NRM (open rectangles) and after 30mT demagnetisation (solid circles) as a function of depth. Grey areas show intervals of anomalous inclination and intensity lows across the plot. (b) Zijderveld diagrams, intensity decay curves and equal-area stereograms extracted from the archive-half section. The stratigraphic position on the section is indicated by grey circles in (a).

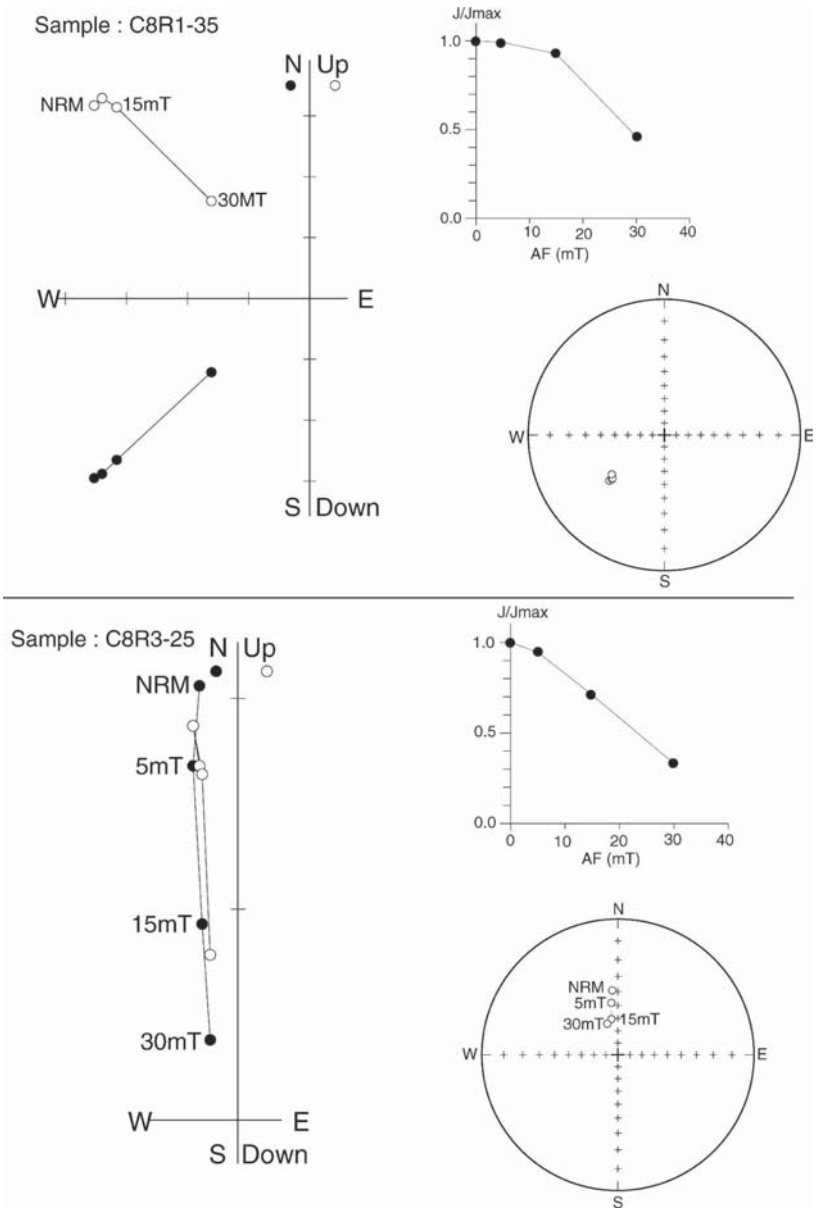


Figure 2 (*Continued*).

Examination of Zijderveld diagrams extracted from two positions in the archive-half long-core measurements revealed two components of magnetisation (Figure 2b). The first is a low coercivity component (LCC) with a lower inclination, often removed by 20 mT. The second is a high coercivity component (HCC) having a steeper inclination and considered to represent the characteristic remanent magnetisation (ChRM) and/or chemical remanent magnetisation (CRM). Strong resistance upon demagnetisation characterises the first component at the upper stable interval, whereas at the lower stable interval, the first and second components have a clear segment with a partially overlapping coercivity spectrum. On the stereographic projection, the inclination changes from lower values for the NRM to progressively steeper inclinations at 30mT (Figure 2b). This remagnetisation might be related to the effect of drilling-induced overprint characterised by downward direction often reported previously (Fuller *et al.* 1998; Acton *et al.* 2002), which would logically result in shallowing of the original NRM inclination prior to demagnetisation.

HOLE 1194B

In hole 1194B a 1.5 m-thick section of basalt was recovered below 421.12 mbsf. The average NRM intensity for the measured interval is 0.48 A/m ($\sigma = 0.27$, $N = 29$). The mean inclination value at NRM is –26.1° ($\alpha_{95} = 5°$, $K = 30.1$, $N = 29$) but this becomes steeper and

reaches a value of –50° on demagnetisation to 30mT (Figure 3). However, at 30mT demagnetisation the inclination becomes consistently uniform with a mean value of –43.6° ($\alpha_{95} = 1.6°$, $K = 290.7$, $N = 29$) representing ChRM identified as normal polarity. We do not have Zijderveld diagrams of measurements extracted from the long core measurements because only two steps of measurements (NRM, 30 mT) were made. Moreover, because the basalt recovered here, in addition to being altered is highly fractured, it was not possible to take adequate samples for further paleomagnetic investigations. Nevertheless, the persistently similar direction after demagnetisation supports the idea that the basalt unit possibly represents a single flow.

HOLE 1197B

Basaltic rocks representing basement in hole 1197B were penetrated at 656 mbsf. The calculated mean NRM intensity from the archive-half long-core measurement is 0.07 A/m ($\sigma = 0.04$); this intensity is low compared with the results from other holes. The remanence magnetisation showed unstable behaviour on demagnetisation, hence inclination values are not consistent and interpretable (Figure 4). However, the measured inclination has positive values indicating a reverse polarity. Before demagnetisation, inclination values are large positive and are more clustered (Figure 4), possibly resulting from the effect of the downward drilling overprint. Zijderveld diagrams from discrete samples (see below) are characterised by well-separated components with distinct segments, although the calculated directions using principal-component analysis were inconsistent. There is no improvement in the clustering of ChRM directions after demagnetisation.

HOLE 1198B

In hole 1198B a 3.5 m section of basalt was recovered below 513.85 mbsf. It is more solid than the material recovered at other sites. As a result, more investigations on discrete samples have been made to support the

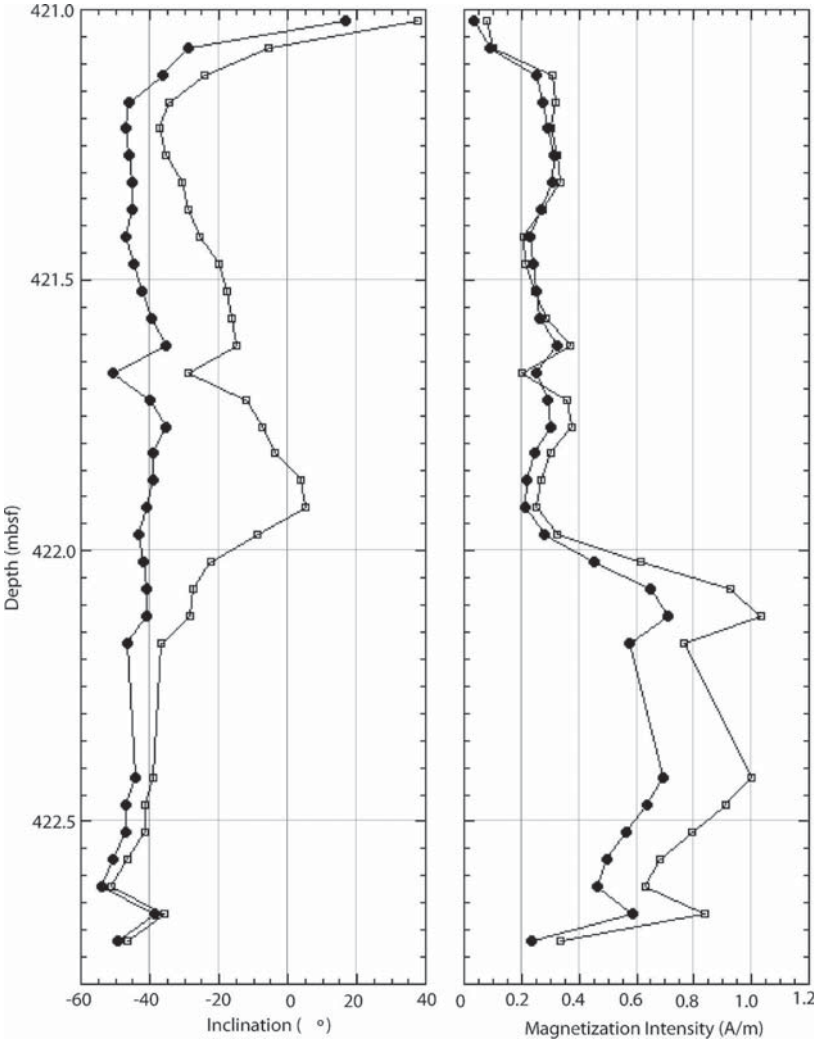


Figure 3 Archive-half measurements on acoustic basement in hole 1194B showing inclination and magnetisation intensity values at NRM (open rectangles) and after 30mT demagnetisation (solid circles) as a function of depth.

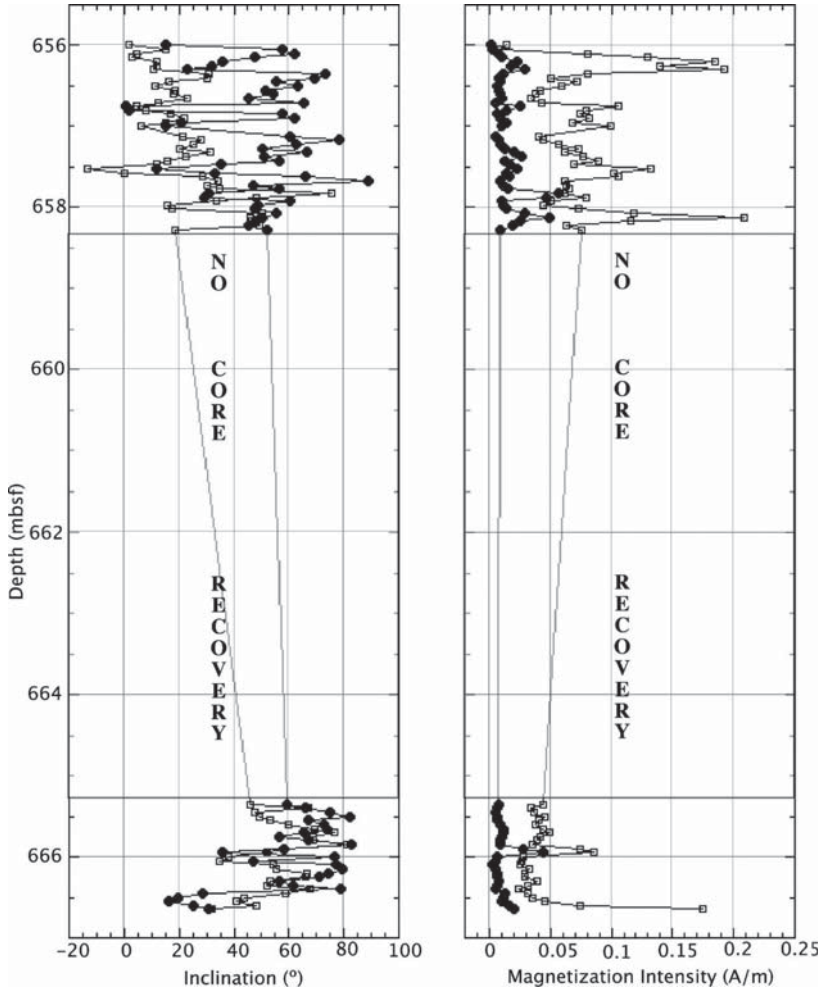


Figure 4 Archive-half measurements on the volcaniclastic breccias in hole 1197B showing inclination and magnetisation intensity values at NRM (open rectangles) and after 30mT demagnetisation (solid circles). The gaps are intervals of non-recovery.

long-core measurements. The long-core measurements yielded a mean intensity of NRM 0.3 A/m ($\sigma = 0.1$) with a mean inclination value of 68.9° ($\alpha_{95} = 2.1°$, $K = 85.6$, $N = 52$). After 30mT demagnetisation the mean inclination becomes 50.6° ($\alpha_{95} = 2.3°$, $K = 77.7$, $N = 52$), significantly lower than the inclination at NRM and represents reverse polarity (Figure 5a). The orthogonal vector component diagrams extracted from two levels in the long-core measurements revealed two components of magnetisation (Figure 5b). The first is a LCC characterised by steep inclination that persists until 20 mT. The second component, which is isolated at 20 mT or beyond, is the HCC and is thought to represent the ChRM.

DISCRETE SAMPLES

Nineteen discrete cube samples have been prepared from the working-half of the recovered basalt samples for additional paleomagnetic study, in order to augment the continuous long-core measurements from holes 1197B and 1198B. Both AF and thermal demagnetisation techniques were used to detect any spurious remagnetisation that might be present. The directional analyses, the Fisher statistics of the discrete samples, and the average of the combined long-core and discrete samples were all carried out by using the PaleoMac (Cogné 2003) software package.

Both thermal and AF demagnetisations undertaken on the samples from hole 1198B revealed two components of magnetisation. The softer component is a poorly defined, but steep, component with a low unblocking temperature (LUTC), while the harder is a well-defined high unblocking temperature component (HUTC), mostly isolated in the temperature range of 500–630°C (Figure 6). Titanomagnetite is the most probable magnetic mineral carrier of LUTC, whereas magnetite and titanohematite are thought to be the probable carriers of HUTC (Figure 6). The AF demagnetisation, like the thermal technique, demonstrated two components of magnetisation—LCC and HCC—mostly with partially overlapping coercivity spectra. In samples where these two components are overlapping, the inclination of the LCC was often isolated by 15 mT. These discrete sample analyses indicate that the ChRM in most of the cases is isolated by the 20 mT stage of the stepwise demagnetisation process and the magnetisation vector after the 30 mT demagnetisation step defined

straight-line segments that are directed towards the origin. The paleomagnetic directions from all the discrete samples are determined by principal-component analysis techniques, with the mean inclination of HUTC ($I=53.2^\circ$, $N=8$, $K=372.3$, $\alpha_{95}=2.9^\circ$) and HCC ($I=49.7^\circ$, $N=11$, $K=199.2$, $\alpha_{95}=3.2^\circ$) determined using PaleoMac (Cogné 2003). The LUTC and LCC both have steep inclinations ($>60^\circ$) and are thought to represent drilling-induced remagnetisation. These results suggest that the LCC carrier, possibly titanomagnetite, is more prone to drilling-induced overprint. Therefore, only the HUTC and HCC components are considered in further directional analyses.

DISCUSSION

Paleomagnetic-mean directions

Combined long-core and discrete paleomagnetic measurements performed on board the ship on the acoustic basalt rocks recovered during ODP Leg 194 reveal consistent and well-defined directional results. The discrete sample measurements indicate that demagnetisation behaviour of the basalts is generally good, and results appear to be reliable.

Table 1 gives the paleomagnetic mean inclinations based on both long core and discrete sample measurements. Results from the three holes can be grouped into two similar paleomagnetic ChRM inclinations.

In group 1 the predominant ChRM inclination is determined both by the long-core and discrete measurements. It is therefore comprised of the overall mean from hole 1193C (between depth intervals of 540.96 and 543.97 mbsf), and the overall mean from hole 1198B (HUTC and HCC from long core and discrete measurements in hole 1198B are first averaged) (Table 1, rows 4, 8, 9, 10). The inclination mean (Figure 7) is $Im=52.7^\circ$, $\alpha_{95}=4.0^\circ$, $K=537.0$, $N=4$, $n=129$.

In group 2, which is often affected by the drilling overprint are measurements from the top part of hole 1193C and the mean from site 1194B (Table 1, rows 2, 6). The inclination mean (Figure 7) is $Im=41.6^\circ$, $\alpha_{95}=8.7^\circ$, $K=820$, $N=2$, $n=45$. The result from this group could not be verified from independent discrete measurements, and because it is highly affected by

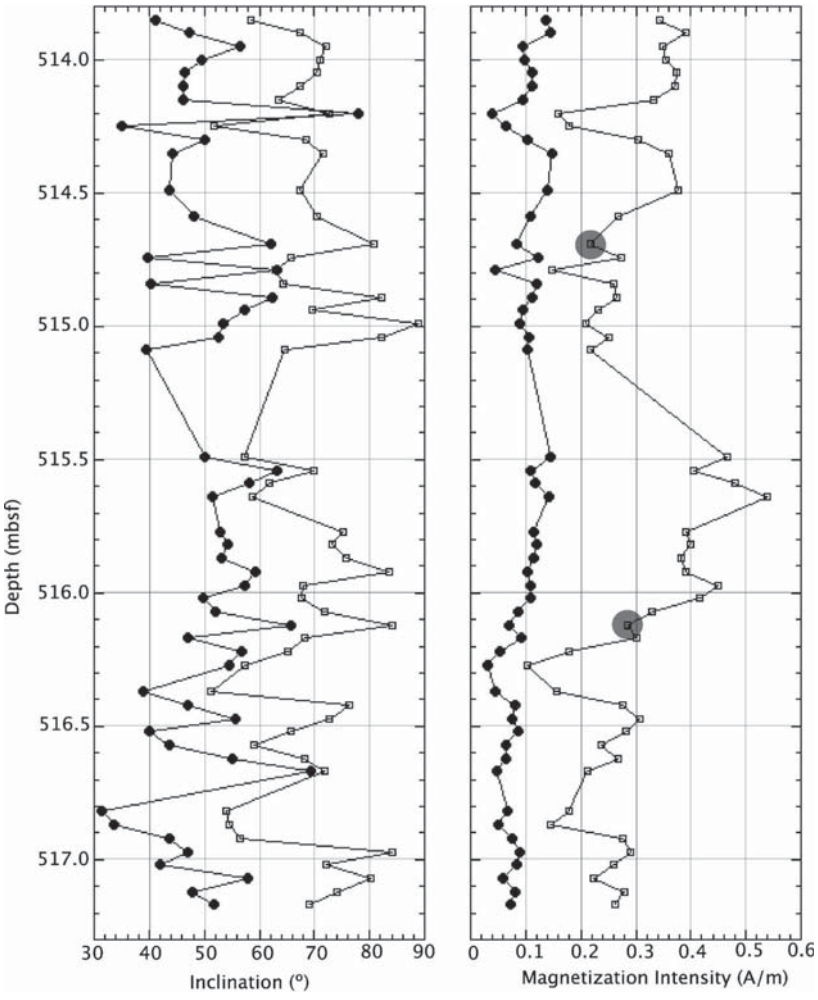


Figure 5 (a) Archive-half measurements on acoustic basement in hole 1198B showing inclination and magnetisation intensity values at NRM (open rectangles) and after 30mT demagnetisation (solid circles) as a function of depth. (b) Zijderveld diagrams together, intensity decay curves and equal-area stereograms extracted from the archive-half section. The stratigraphic position on the section is indicated by grey circles in (a).

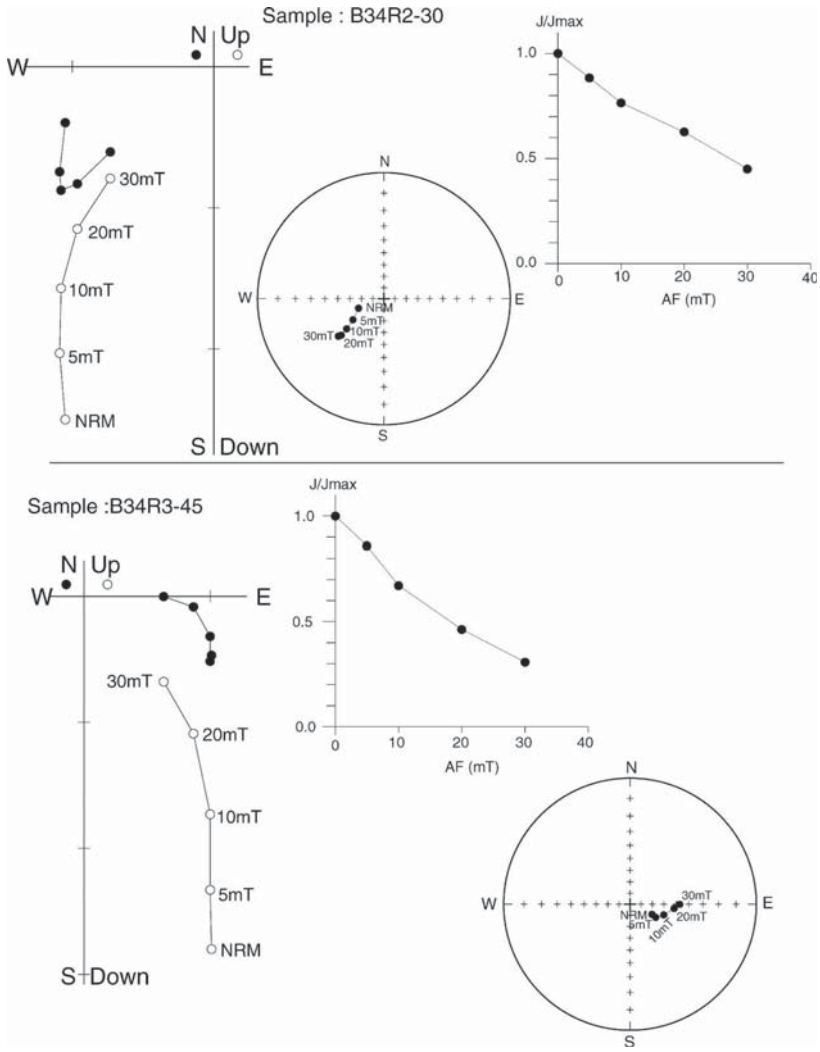


Figure 5 (*Continued*).

drilling overprint, it will not be considered in further interpretations.

Interpretation

Using the average paleomagnetic inclination of group 1, the corresponding paleolatitude is computed to be 33.3°S (with upper and lower 95% confidence limits of 37.3°S and 29.6°S). Comparisons of the paleo-colatitude resulting from this (52.7°–56.7°S–60.4°) with the recent Australian apparent polar wander path (APWP) curve of Besse & Courtillot (2003) (Figure 8) suggest two possible ages: the first between 130 and 110 Ma, the second between 190 and 165 Ma.

On the other hand, the Australian APWP curve of Musgrave (1989) for the last 100 Ma gives the age estimates to be between 85 and 58 Ma. Similarly, the Australian APWP curve determined from the Indian APWP by G. D. Acton (pers. comm. 2002) suggests an age older than 70 Ma. In the absence of declination control for the deep-sea drilling samples and because of the shape of the Australian APWP, a unique age estimate is not possible: two ages are possible. However, the $^{40}Ar/^{39}Ar$ age determination on one of the recovered basalts gave an age of 162 ± 0.9 Ma (Karner *et al.* 2002), which suggests that the second possible age estimate (190–165 Ma) given by the APWP of Besse & Courtillot (2003) is the more likely. Therefore, our results suggest that the time of olivine basalt emplacement could be around 160 Ma. Geochemical analyses on the basalt (Karner *et al.* 2002) showed that it is characterised by an extremely low Ca/K ratio possibly indicating non-oceanic (i.e. non-MORB) origin of the basalt. Further, these results together with inspection of the cored basalt suggest that the area was probably subaerial during the emplacement and could represent a syn-rift emplacement when basin formation began around the Townsville Trough, during Middle–Late Jurassic (Struckmeyer & Symonds 1997; Karner *et al.* 2002).

Regional lateritisation is a well-known phenomenon in Australia during Late Oligocene to Early Miocene times which resulted in chemical remagnetisation affecting formations with ages ranging from Late Paleozoic to Mesozoic strata (Schmidt & Embleton

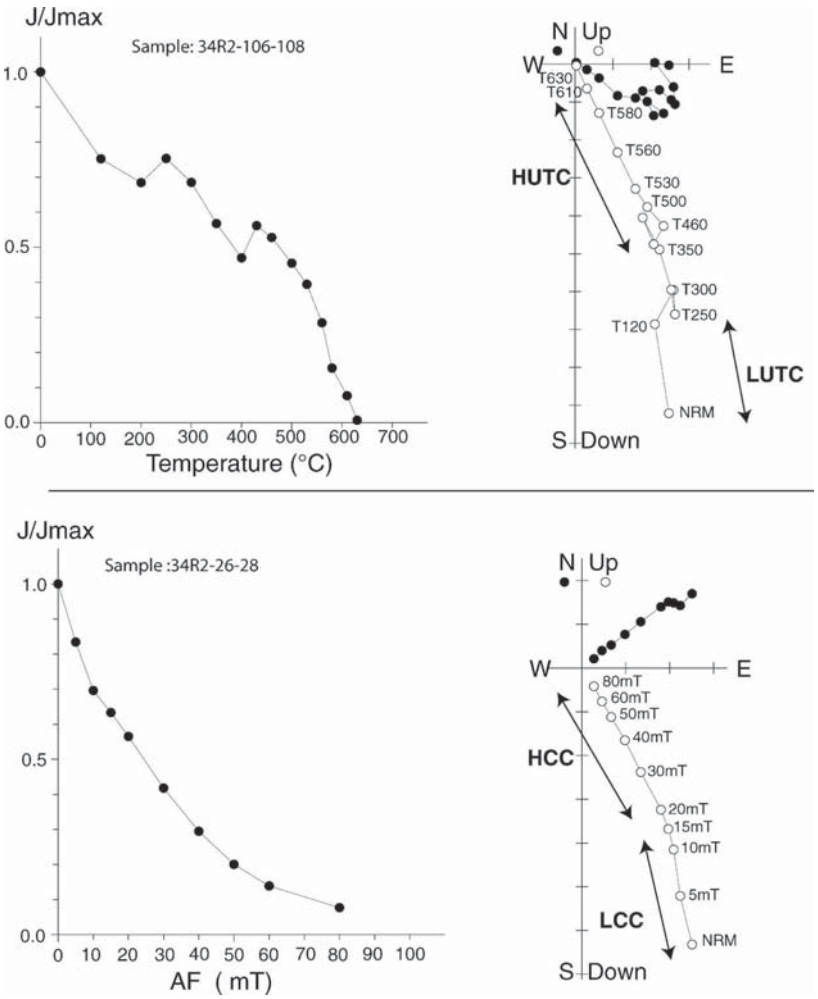


Figure 6 Normalised intensity curves and Zijderveld diagrams for typical discrete specimens from the working-half of hole 1198B: (top) thermal (Th) demagnetisation (temperature in °C) and (bottom) alternating (AF) field (field in mT) demagnetisation.

Table 1 Summary of the calculated inclination values from various holes and discrete samples.

Hole	Coordinates	Section[a]	Depth intervals (mbsf)	Sample type[b]	Measurement[c]		n[d]	I[e]	k[f]	α_{95}[g]
					Type	Demagnetisation				
1. 1193C	20.4°S, 151.7°E	8R1 (25–105)	539.15–539.95	C	NRM	0	16	−39.6	40.8	5.8
2. 1193C	20.4°S, 151.7°E	8R1 (25–105)	539.15–539.95	C	AFD	30 mT	16	−41.7	38.2	6.1
3. 1193C	20.4°S, 151.7°E	8R2 80–8R490	540.96–543.97	C	NRM	0	58	−43.8	126.2	1.7
4. 1193C	20.4°S, 151.7°E	8R2 80–8R490	540.96–543.97	C	AFD	30 mT	58	−57.5	143.9	1.6
5. 1194B	20.4°S, 152.0°E	34R4 20–34R535	421.12–422.72	C	NRM	0	29	−26.1	30.1	5.0
6. 1194B	20.4°S, 152.0°E	34R4 20–34R535	421.12–422.72	C	AFD	30 mT	29	−43.6	290.7	1.6
7. 1198B	21.1°S, 152.7°E	34R185–34R3145	513.85–517.17	C	NRM	0	52	68.9	85.6	2.1
8. 1198B	21.1°S, 152.7°E	34R185–34R3145	513.85–517.17	C	AFD	30 mT	52	50.6	77.7	2.3
9. 1198B	21.1°S, 152.7°E	34R1–34R4	513.85–517.17	D	AFD	0–80 mT	11	49.7	199.2	3.2
10. 1198B	21.1°S, 152.7°E	34R1–34R4	513.85–517.17	D	THD	0–630°C	8	53.2	372.3	2.9
Means			Rows considered for mean				N[d]	I[e]	K[f]	α_{95}[g]
Group 1 mean			Rows 4, 8, 9, 10				4	52.7	537	4.0
Group 2 mean			Rows 2, 6				2	41.6	820	8.7

[a]Section and type and number of cores with location interval ranges given in centimetres.
[b]C, archive-half long core; D, discrete.
[c]AFD, alternating field demagnetisation (mT); NRM, natural remanent magnetisation; THD, thermal demagnetisation (°C).
[d]n, number of samples or measurements; N, number of site means.
[e]I, site mean inclination in stratigraphic coordinates and overall inclination.
[f]k/K, Fisher precision parameter at site means and at overall mean.
[g]α_{95}, 95% confidence interval at site means and at overall mean.

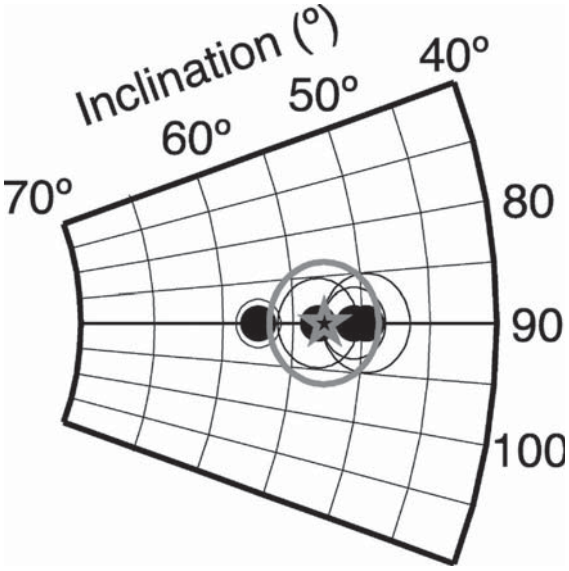


Figure 7 Stereographic projection showing overall mean calculation for the inclinations observed from the archive-half and discrete samples (those treated in AF and Th). The four full black circles give the inclination means (rows 4, 8, 9, 10 in Table 1), while the star symbol shows the overall mean direction (group 1 mean in Table 1) for the acoustic basement of the Marion Plateau.

1976). A detailed paleomagnetic study by Acton & Kettles (1996) in the New England Orogen has identified lateritisation effects on Tertiary volcanics with age estimates ranging between 26–18 Ma and 56–6 Ma for volcanics from the central volcanic province. Our observation at hole 1193C of two stable polarities separated by two excursions on a single weathered lava flow may be considered as indications of lateritisation and weathering. However, due to the known overprinting problems related to the drilling and the difficulties of discriminating this phenomenon from other low-stability remagnetisation components, interpretations on the possible ages of remagnetisation would be speculative.

CONCLUSIONS

Results of paleomagnetic investigations on combined long-core and discrete samples collected from the basaltic rocks recovered from ODP Leg 194 showed strong magnetisation intensity that provided good and reliable directional results. The isolated vector component diagrams showed two components of magnetisation, with magnetic mineralogy not difficult to establish.

The well-behaved directions from the long core measurements at hole 1193C (540.97–543.97 mbsf) and from the long core and discrete measurements at hole

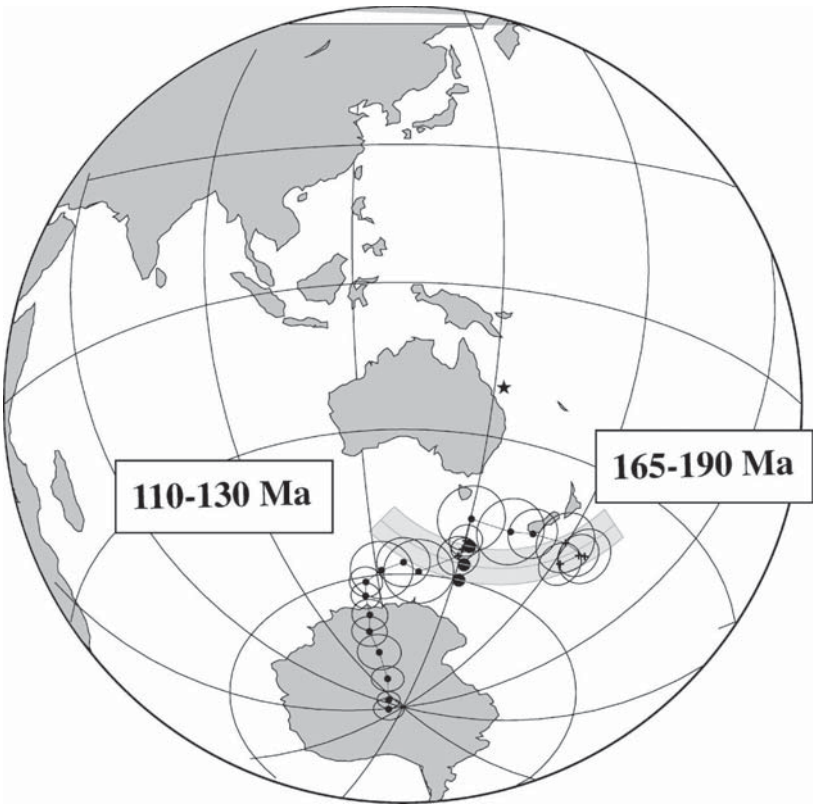


Figure 8 The observed paleolatitude (the centre large black dot) with their error margins (the two outer large black dots) are plotted on the Besse & Courtillot (2003) Australian APWP as arcs of small circles about the latitude of the site for age estimate (dashed grey arcs). The five crosses of the APWP within the dashed grey arcuate area are considered to be possible ages for the acoustic basement.

1198B (513.85–517.25 mbsf) were used to calculate the overall mean inclination. This yielded an overall mean inclination value of 52.7° ($\alpha_{95} = 4.0°$, $K = 537.0$, $n = 129$, $N = 4$). From this result the paleolatitude of the Marion Plateau during the emplacement of the basalt is 33.3°S (with upper and lower 95% confidence limit 37.3°S and 29.6°S, respectively). The paleo-colatitude is given by, 90° paleolatitude, which would be 56.7°S with upper and lower 95% confidence limit 60.4°S and 52.7°S, respectively. When these paleo-colatitudes are compared with the Australian APWP, a window of ages between 190 and 165 Ma is the most likely time of emplacement of the basalt. This age is confirmed by the $^{40}Ar/^{39}Ar$ age of 162 ± 0.9 Ma obtained for this basalt by Karner *et al.* (2002).

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